

Q-Folio: Constraint-Preserving QAOA for Portfolio Selection

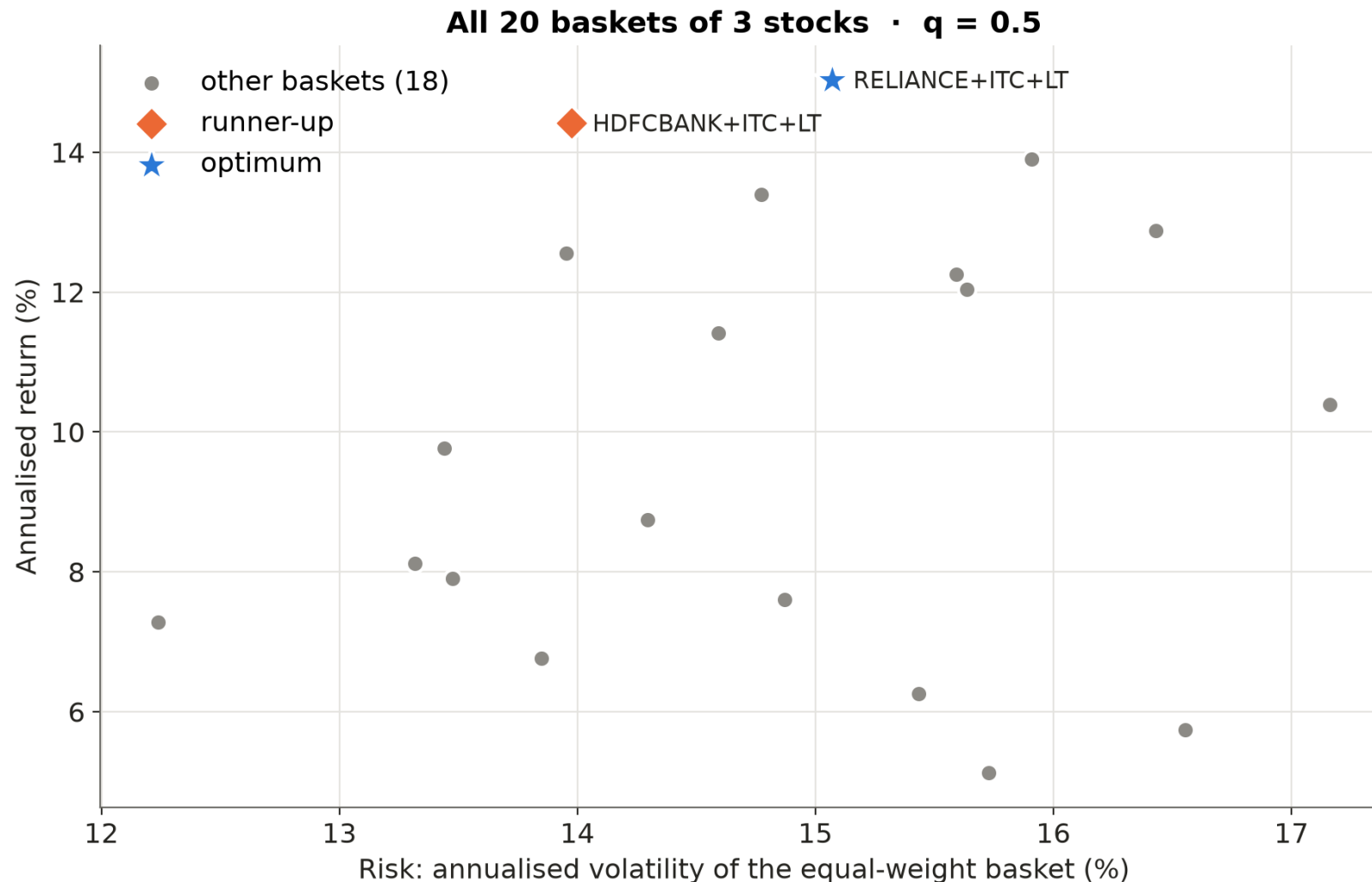
Qiskit Fall Fest 2026 · Industry Track I2

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[Real results on IBM Quantum ibm_kingston](#)

Picking the best 3 of 6 stocks, on a quantum computer

Robo-advisor model baskets · NSE large caps · return $\uparrow$ risk $\downarrow$



All $C(6,3) = 20$ baskets on the risk/return plane.

Basket count grows as $C(n,k)$:
~185 000 for 10-of-20.

Goal: find the best basket for a client's risk aversion q .

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Put the constraint in the circuit, not in the cost

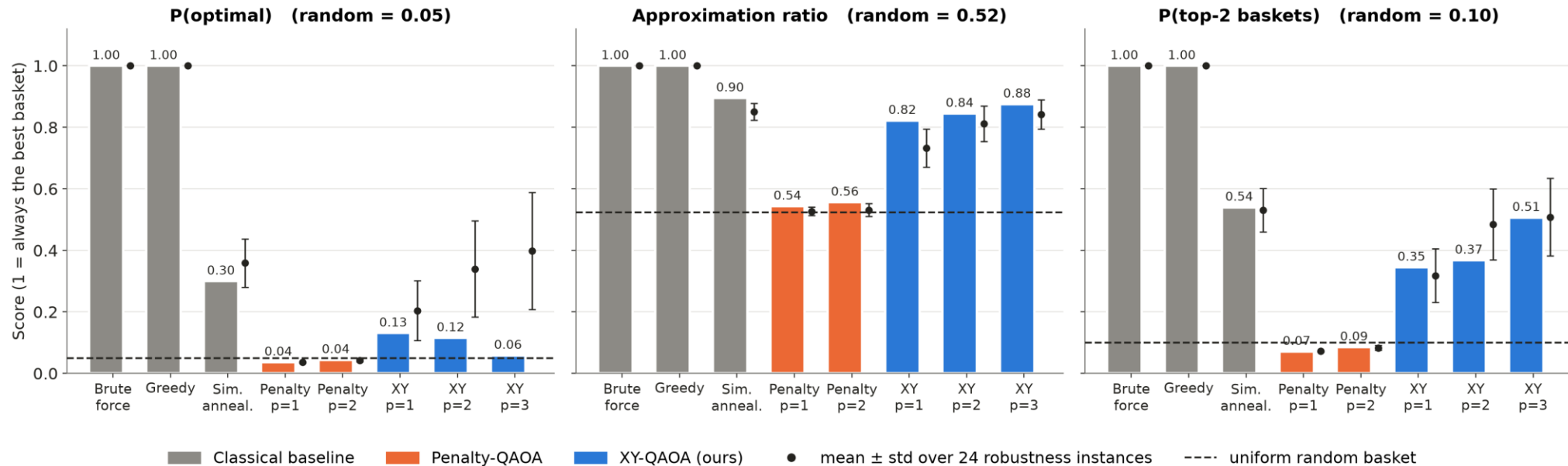
Penalty QAOA (X mixer)

Searches all 64 bit strings, 44 invalid
→ finds the best basket at chance level

XY-QAOA (Dicke state + XY mixer)

Only ever visits the 20 valid baskets
→ every shot is a real portfolio

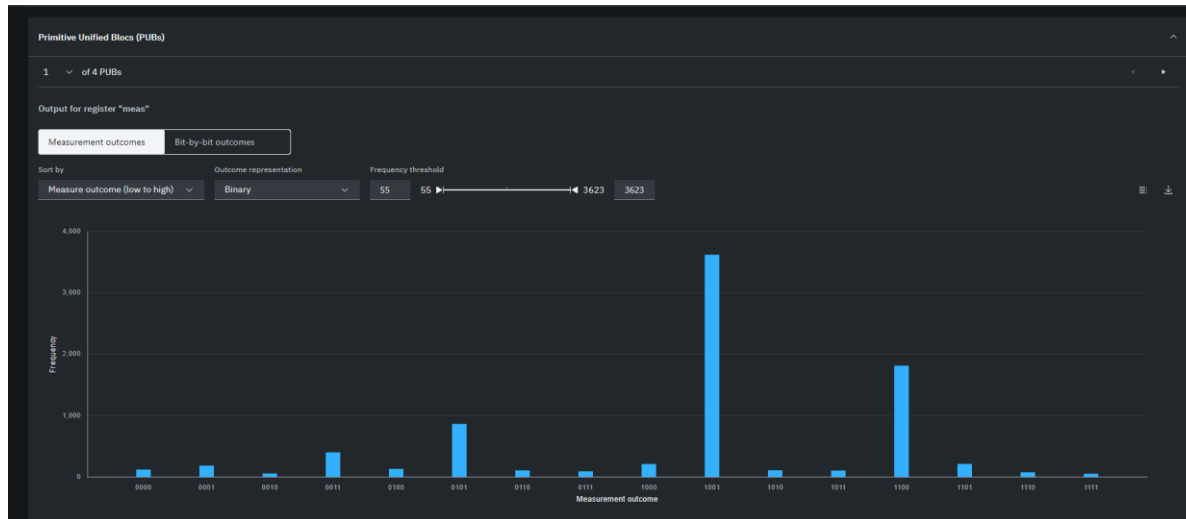
Headline instance: pick 3 of 6 NSE stocks (RELIANCE, TCS, HDFCBANK, INFY, ITC, LT), $q = 0.5$ · ideal simulator



100% valid baskets · P(optimal) 0.34 vs 0.05 random
comparable to simulated annealing · penalty-QAOA = random

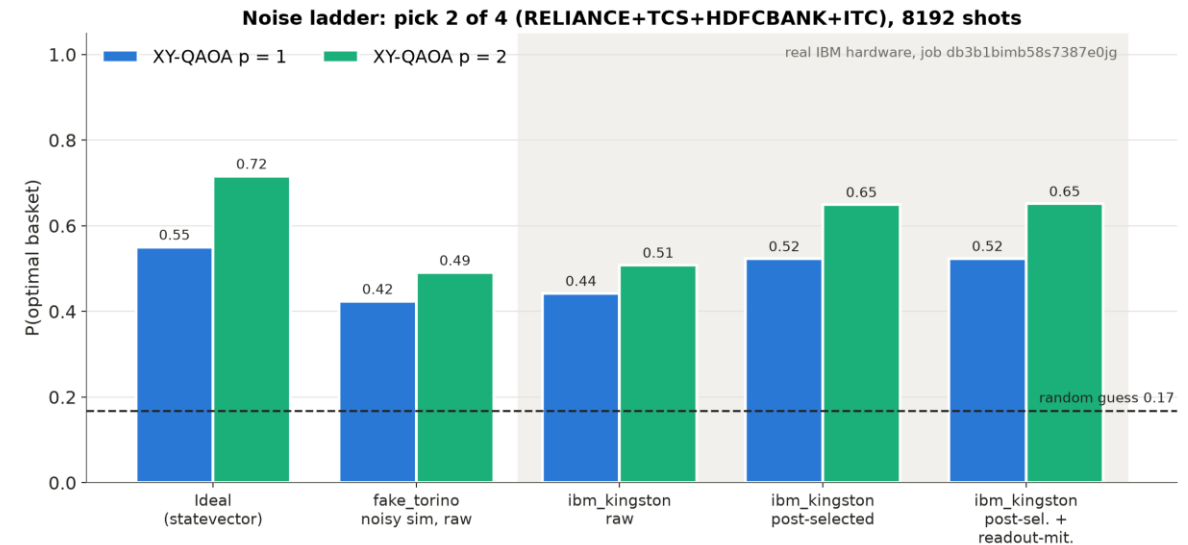
Real IBM quantum hardware: ibm_kingston

ibm_kingston: 0.51 raw $\rightarrow$ 0.65 post-selected (random 0.17); top answer = true optimum



Real ibm_kingston output — tallest bar 1001 = optimal basket (RELIANCE+ITC), second 1100 = runner-up

Limits: brute force is still faster at this size, so no quantum advantage is claimed; today's noise limits us to ~4–6 assets.



Live demo



Repo